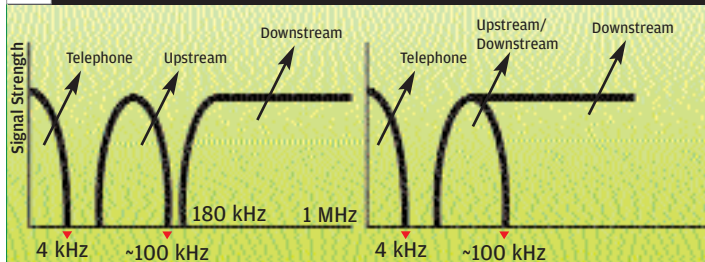


Frequency Splitting In FDM And Echo Cancellation



In FDM (left), the frequency band is split into ranges for telephone service, upstream data, and downstream data, at the frequencies indicated above. In Echo Cancellation (right), the frequency bands for upstream and downstream data overlap

Phone service is at the lower end of the spectrum; data is at the higher end. To implement this frequency splitting, you need to install a splitter along with the DSL modem. Voice can typically be transmitted within the 4 KHz range.

The actual splitting of the frequency range is done using one of two methods—*Frequency Division Multiplexing (FDM)* or *Echo Cancellation*. FDM uses separate frequency bands for the transmitting and receiving channels. In Echo Cancellation, transmission and reception happens on the same frequencies—that is, the bands overlap.

The advantage of FDM is that it is cheaper to implement in the sense that the hardware needs to do less work to separate the transmitted and received signals; the advantage of Echo Cancellation is that it transmits at a relatively lower end of the spectrum, meaning less signal attenuation with distance. See figure *Frequency Splitting In FDM And Echo Cancellation*.

Actually, it's not quite as simple as that. There's modulation involved as well (refer this month's *Fast Track* for more on modulation). Modulation is required because of several factors, including the noise associated with transmitting data over the kinds of frequencies we're talking about. In ADSL, the signal from your home is modulated (most usually) using a technique called *discrete multi-tone (DMT)*, which is also what has been approved by the standards authorities for VDSL.

DMT splits up the spectrum on which the downstream signal travels into 247 channels, each of these 4 KHz wide (and the downstream spectrum into 32 channels). The biggest advantage of splitting up the band into so many channels is the continuous quality check that is possible.

Each of the hundreds of channels is monitored, and if and when it is seen that quality is impaired on a particular channel, the data on that channel can be shifted onto another one. This happens continuously—the system is constantly re-assigning data to channels. What results is the "best" channels at any particular moment being chosen.

After all the data is switched onto the right place, your phone line transmits everything



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